

MOHAMMED AFFAAN KHAN

GenAI & Agentic AI Engineer · Voice Intelligence · RAG Systems · Full-Stack AI

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https://affaanh.github.io/Mohammed_Affaan_Khan_portfolio/

PROFESSIONAL SUMMARY

Agentic AI Engineer with 2+ years of hands-on experience designing and shipping production-grade conversational voice agents, RAG-powered knowledge systems, and multimodal AI platforms across MedTech, FinTech, and SaaS verticals. Specializes in end-to-end AI pipeline architecture — from speech ingestion and LLM reasoning to avatar animation, automated workflow orchestration, and ML-driven recommendation engines — delivering <500ms latency at 10,000+ concurrent call scale. Demonstrated track record cutting clinical reporting time by 85% and operational costs by 45% through intelligent automation. Deep expertise across the modern agentic stack: LangChain, LangGraph, CrewAI, n8n, Millis.ai, Pinecone, pgvector, ElevenLabs, PyTorch, and multi-cloud LLMs (GPT-4o, Claude 3.5, Llama 3). Proven ability to own full project life cycles — from architecture through deployment and MLOps — as a high-ownership individual contributor in fast-moving startup environments.

IMPACT AT A GLANCE

<500ms	10,000+	99.9%	85%	45%
Voice Latency	Concurrent Calls	SLA Uptime	Faster Reporting	Cost Reduction

PROFESSIONAL EXPERIENCE

Maskan Technologies

May 2025 – Present

AI Developer · Bangalore, India

- **Conversational AI Voice Agent Platform** Millis.ai · Whisper STT · GPT-4o · LangChain · ElevenLabs · Kubernetes
 - Architected end-to-end real-time voice intelligence platform on Millis.ai, sustaining **10,000+ concurrent calls** with **99.9% uptime SLA** and **<500ms** total STT→LLM→TTS latency in production.
 - Engineered agentic decision loop via LangChain with persistent multi-turn context memory and dynamic role-switching; automated **2,000+ daily interactions**, reducing operational costs by **45%**.
 - Integrated n8n workflow automation for post-call orchestration — CRM updates, appointment scheduling, and escalation routing — eliminating manual intervention across **100%** of routine interactions.
 - Deployed Kubernetes auto-scaling policies absorbing **10x traffic spikes**; cut model inference costs by **35%** via INT8 quantization and batched GPU utilization.
- **Real-Time AI Avatar with Retrieval-Augmented Generation** Pinecone · GPT-4o · LlamaIndex · ElevenLabs TTS · Facial Animation Engine
 - Built multimodal digital human delivering **200–400ms** response latency through a synchronized RAG pipeline, neural TTS, and real-time facial animation engine deployed to live production audiences.
 - Indexed **200+ hours** of structured content into 768-dimension Pinecone vector embeddings; engineered self-updating knowledge pipeline with automatic re-indexing on source changes.
 - Applied voice cloning and emotion-adaptive prosody modeling to produce context-aware, lifelike responses — measurably improving pilot engagement scores.
- **NHS AI-Powered Clinical Workflow System** U-Net · PyTorch · VTK · FastAPI · PostgreSQL · React · DICOM
 - Engineered deep learning CT scan analysis pipeline using U-Net segmentation achieving **Dice coefficient >0.85** and hybrid malignancy classifier at **92% accuracy** for lung nodule risk stratification.
 - Developed full role-based clinical workflow platform (Pulmonologist · Radiologist · Admin) covering referral management, nodule tracking, MDT discussions, and automated clinical decision logging across RESTful APIs.
 - Built automated 2D/3D CT visualization suite with VTK-based MPR/Glassy viewer; slashed radiologist reporting time from **30 min to ~5 min (85% reduction)** via auto-generated PDF/MP4 reports, with pilot data showing **40%** workflow efficiency gain.

► **Xplortiv – AI Recommendation & Discovery Engine** PyTorch · BGE Embeddings · pgvector/AlloyDB · BigQuery · Vertex AI · GRU

- ❑ **Architected** multi-phase booking-conversion-optimized ranking system: GRU-based short/medium/long-term event encoder with task-conditioned gating across 9 prediction heads (skip, watch, book, verify, capture) with prioritized loss weights (booking: 3.0–4.0, verified attendance: 5.0).
- ❑ **Designed** semantic search engine for natural-language venue discovery ("date night in Glasgow", "rooftop bar nearby") via BGE-small-en 128-D embeddings served through pgvector/ScaNN on AlloyDB for real-time sub-50ms retrieval.
- ❑ **Implemented** verified-content weighting system: captures from QR-verified attendees receive a trust multiplier, creating a defensible ranking signal unavailable to social platforms without a booking + verification layer.
- ❑ **Built** end-to-end MLOps pipeline: BigQuery for offline training data assembly, Vertex AI for model training and registry, AlloyDB for per-user embedding snapshots, and INT8 quantization for production inference deployment.

► **Maskanza – Multi-Tenant Restaurant Management SaaS** Node.js · Express · PostgreSQL · React · Socket.IO · Stripe · i18next

- ❑ **Designed and shipped** production-grade multi-tenant SaaS encompassing POS, Kitchen Display System, QR ordering, inventory management, reservations, and analytics across **132 API endpoints** and **53 frontend pages**.
- ❑ **Implemented** real-time order orchestration via Socket.IO, Stripe subscription billing with webhook lifecycle management, and conversational AI booking/ordering agent — with **9-language i18n** support including RTL Arabic and 23 RBAC permission scopes.

► **OpenMontage – Agentic AI Video Production System** LangChain · GPT-4o · ElevenLabs · Remotion · Pinecone · FAISS

- ❑ **Architected** instruction-driven agentic video production pipeline with 11 specialized pipeline types (explainer, cinematic, avatar-spokesperson, localization-dub) orchestrated by an LLM agent via declarative YAML manifests and director skills.
- ❑ **Built** dynamic tool registry enabling zero-code provider swapping, capability discovery, and cost governance across multi-provider AI asset generation (video, image, TTS, music) with Remotion and HyperFrames composition runtimes.

► **SiteChat RAG Bot & Estimate256 ML Estimation Engine** FAISS · OpenRouter · FastAPI · XGBoost · Scikit-learn · Alembic · React

- ❑ **Built** SiteChat: RAG-powered website chatbot platform — URL discovery → FAISS vector indexing → GPT-4o chat — deployed as a FastAPI + React SPA enabling zero-code bot creation for any website.
- ❑ **Developed** Estimate256: production ML estimation engine with three domain-specific XGBoost models, FastAPI backend, catalog/billing/auth layer, and Alembic-managed PostgreSQL schema migrations.

Software Engineering Intern — JP MORGAN CHASE & CO.

Sep 2023 – Dec 2024

Software Engineer Intern · Remote

- ❑ **Built** 5 production web applications using FastAPI and React; developed Python data processing pipelines (Pandas, NumPy) for large-scale analytical datasets.
- ❑ **Contributed** to MLOps pipelines including model versioning, automated retraining workflows, and A/B testing frameworks; resolved critical bugs reducing incident backlog in legacy production systems.

BYJU'S

Jan 2023 – Jul 2023

Business Development Associate · Bangalore, India

- ❑ **Engaged 500+ prospective clients**, surfacing product requirements that directly informed EdTech AI feature roadmap prioritization; developed executive stakeholder communication and consultative sales skills.

TECHNICAL SKILLS

LLM & Agentic AI	GPT-4o, Claude 3.5, Llama 3, LangChain, LlamaIndex, LangGraph, CrewAI, AutoGen, RAG, Multi-Agent Systems, Prompt Engineering, RLHF, Fine-tuning
Voice & NLU	Millis.ai, ElevenLabs, Whisper STT, Google TTS, WebRTC, SIP/VoIP, Voice Cloning, Emotion-Adaptive TTS, Intent Classification
ML / Deep Learning	PyTorch, TensorFlow, HuggingFace Transformers, U-Net, YOLOv5, XGBoost, Scikit-learn, GANs, GRU, Contrastive Learning, INT8 Quantization
Vector DB & Search	Pinecone, pgvector, AlloyDB (ScaNN), FAISS, Weaviate, Qdrant, Semantic Search, Embedding Pipelines, BGE Embeddings
Workflow Automation	n8n, Zapier, Make, LangGraph, CrewAI — CRM automation, post-call orchestration, multi-step agentic workflows

Computer Vision	OpenCV, VTK, SimpleITK, DICOM Processing, 3D Rendering, MPR Visualization, YOLOv5 Defect Detection
Backend & APIs	Python, FastAPI, Node.js, Express, REST APIs, Socket.IO, PostgreSQL, Sequelize, Alembic, Supabase
Frontend	React, Vite, Tailwind CSS, i18next (RTL/LTR), Stripe SDK, Socket.IO client
MLOps & Infra	Kubernetes, Docker, CI/CD, Google Vertex AI, BigQuery, Model Registry, A/B Testing, GPU Optimization, INT8 Quantization

EDUCATION

B.Tech – Mechanical Engineering (AI/ML Electives · Computer Vision · Data Structures) 2023
Reva University, Bangalore, India

Capstone Project: YOLOv5-based Infection Detection System — achieved **95% defect detection accuracy**; internally published at Reva University.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- ❑ **Self-directed specialization** in Agentic AI Systems, Voice Pipeline Engineering (Millis.ai, ElevenLabs), Workflow Automation (n8n, LangGraph, CrewAI), and Vector Database Architecture
- ❑ **AI/ML Electives** — Reva University: Computational Mechanics, Computer Vision, Data Structures & Algorithms
- ❑ **Capstone Recognition:** YOLOv5 Infection Detection System — 95% defect detection accuracy, internally published at Reva University

Languages: English (Native) · Hindi (Native) · Urdu (Conversational)